

First-Principles Prediction of a Room-Temperature Ferromagnetic Janus VSSe Monolayer with Piezoelectricity, Ferroelasticity, and Large Valley Polarization

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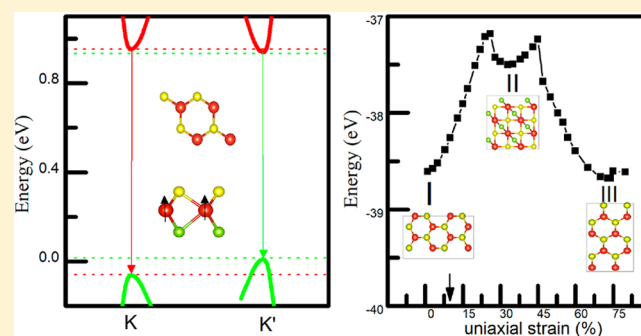
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Supporting Information

ABSTRACT: Inspired by recent experiments on the successful fabrication of monolayer Janus transition-metal dichalcogenides [Lu, A.-Y.; et al. *Nat. Nanotechnol.* 2017, 12, (8), 744 and ferromagnetic VSe₂ [Bonilla, M.; et al. *Nat. Nanotechnol.* 2018, 13, (4), 289], we predict a highly stable room-temperature ferromagnetic Janus monolayer (VSSe) by density functional theory methods and further confirmed the stability by a global minimum search with the particle-swarm optimization method. The VSSe monolayer exhibits a large valley polarization due to the broken space- and time-reversal symmetry. Moreover, its low symmetry C_{3v} point group results in giant in-plane piezoelectric polarization. Most interestingly, a strain-driven 90° lattice rotation is found in the magnetic VSSe monolayer with an extremely high reversal strain (73%), indicating an intrinsic ferroelasticity. The combination of piezoelectricity and valley polarization make magnetic 2D Janus VSSe a tantalizing material for potential applications in nanoelectronics, optoelectronics, and valleytronics.

KEYWORDS: VSSe, piezoelectric, valley, ferroelasticity, ferromagnetic



The technological aspiration of miniaturizing combined multifunctional electronics and information storage devices makes the integration of multiple electronic functionalities in one nanomaterial a highly sought-after target.^{1–3} Two-dimensional (2D) multiferroics exhibiting more than two ferroic properties, ferromagnetism, ferroelasticity, and ferroelectricity,⁴ have been proposed as a material platform to replace silicon in electronics.⁵ Additionally, electromechanical-coupled devices that are piezoelectric have also recently attracted great research interests, as they exhibit switchable electrical polarization under mechanical strain. These have potential applications in sensors, power generation, and electronics.^{6–8} In order to search for compounds that couple magnetic, mechanical, and electric responses, great efforts have been devoted in studying geometrical symmetries that lead to novel material properties in the presence of external stimuli. Generally, materials without central symmetry possess intrinsic polarization domains⁹ and are piezoelectric; for example, this includes ZnO, GaN, and transition-metal dichalcogenides (TMD).¹⁰ Materials with spontaneous deformation or magnetization can be switchable by external strain or magnetic field and are ferroelastic or ferromagnetic.⁴ However, rarely 2D materials have both intrinsic ferroelasticity and ferromagnetism, and those coupling magnetic, mechanical, and electric interactions are even rarer. So far, there are no reported 2D

materials that exhibit both intrinsic ferroelasticity and ferromagnetism at room temperature.

Recently, the experimentally synthesized Janus TMD monolayers⁶ have attracted great interest, because they can harbor both in-plane and mirror asymmetry. Janus TMD monolayers not only preserve the valley properties (inequivalent valley Berry curvature) when compared to conventional TMD systems but also possess Rashba splitting, a spontaneous out-of-plane dipole, as well as a large intrinsic piezoelectric effect.⁶ However, magnetism is absent in all the reported Janus systems, namely, in MoSSe, WSSe, WSeTe, and WTe.¹¹ In these compounds, the energy valley is protected by the orbital magnetic moment, i.e., by time-reversal symmetry.¹² In Janus materials, an opposite Berry curvature is generated as the inversion symmetry is broken, while an external magnetic field¹³ or spontaneous magnetization¹⁴ could lead to the violation of time-reversal symmetry, thus lifting the valley degeneration. A magnetic Janus monolayer fully fulfils these requirements. Most recently, 2D magnetic VSe₂ was successfully fabricated in the laboratory.¹⁵ One open question is whether we can introduce broken mirror symmetry into

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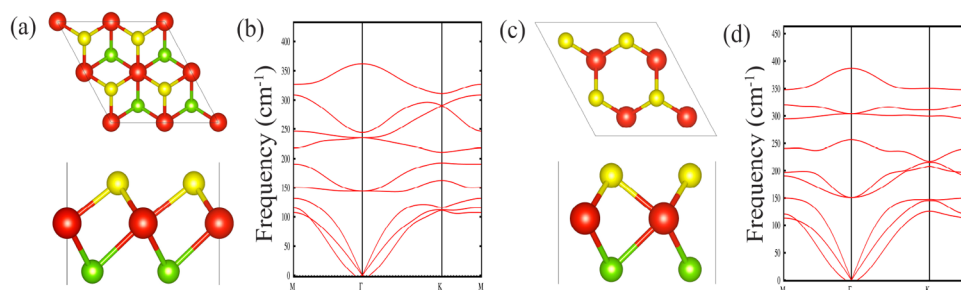


Figure 1. Atomic structures in a $2 \times 2 \times 1$ supercell and the phonon spectrum for the (a,b) 1T and (c,d) 2H VSSe monolayers. Yellow, red, and green spheres represent S, V, and Se atoms, respectively.

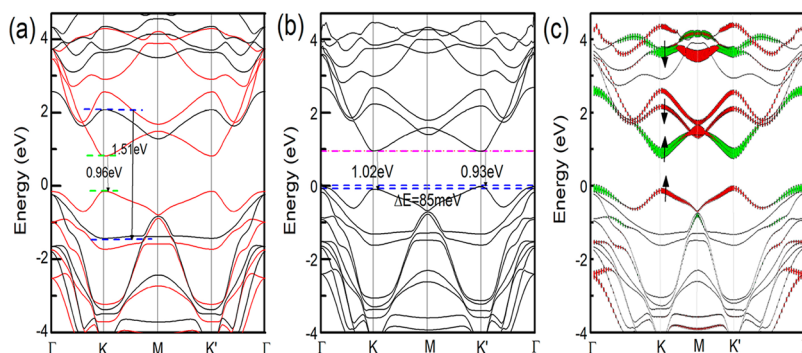


Figure 2. Spin-polarized band structures of the 2H magnetic Janus VSSe monolayer calculated (a) without and (b) with SOC by HSE. The red and black lines in Figure 2a represent the spin-up and spin-down band structures. Figure 2c presents the orbital-resolved band structure calculated with SOC, and the red circle denotes contributions from the vanadium $d_{xy}/d_{x^2-y^2}$ orbitals, while the green circles are for the d_z orbital. As it can be seen in Figure 2c, black arrows with up and down directions represent bands from spin-up and spin-down electrons, respectively. Thus, four bands are arising from the vanadium spin-up $d_{xy}/d_{x^2-y^2}$ orbital, spin-up d_z orbital, spin-down $d_{xy}/d_{x^2-y^2}$ orbital, and spin-down d_z orbital, respectively. The Fermi level is set to zero in every figure.

VSe₂ by designing a magnetic Janus VSSe monolayer? Magnetic VSSe might be an excellent multifunctional material if proved to be energetically and dynamically stable.

In this work, we predict for the first time a new room-temperature ferromagnetic Janus VSSe monolayer by first-principles approaches and particle-swarm optimization (PSO). 1T and 2H ferromagnetic Janus VSSe monolayers are first identified to be the most stable phases. Time-reversal symmetry is violated by an intrinsic magnetic field, which originates from an exchange interaction between the vanadium $d_{x^2-y^2}/d_{xy}$ and d_z orbitals. A large direct valley polarization (~ 85 meV) is thus predicted due to the combination of spin-orbit coupling (SOC) and exchange interaction and further validated by a $k \cdot p$ model. When compared to a VSe₂ monolayer, in which the polarization is forbidden due to mirror symmetry, the magnetic Janus VSSe monolayer possesses an intrinsic out-of-plane dipole moment, leading to a large vertical piezoelectric polarization. When Janus VSSe undergoes uniaxial in-plane strain, a giant in-plane piezoelectric effect (3.303×10^{-10} C/m) is generated. Remarkably, such novel ferroelasticity is demonstrated in the magnetic Janus VSSe monolayer with an ultrahigh reversible strain up to 73%. This is much higher than that reported in other 2D materials,¹⁶ indicating an extremely strong switching signal.

The structural search for 2D magnetic VSSe in the ground state was carried out by using the particle swap method as implemented in the CALYPSO code.¹⁷ The electronic structure calculations were performed by using density functional theory (DFT) within the generalized gradient approximation of the Perdew–Burke–Ernzerhof (PBE) func-

tional by the Vienna ab initio simulation package (VASP).^{18–20} A hybrid functional based on the Heyd–Scuseria–Ernzerhof (HSE) exchange–correlation functional²¹ was adopted for accurately calculating band structure. A long-range van der Waals interaction (DFT-D3 method) was incorporated to correct total energy.²² A vacuum layer with a thickness of 15 Å was used to avoid artificial interactions between neighboring layers, and an energy cutoff of 500 eV was set for the plane-wave basis set. The monolayer VSSe was relaxed until the energy and the forces were converged to 10^{-6} eV and 0.001 eV/Å, respectively. The Monkhorst–Pack k -point meshes used are $19 \times 19 \times 1$ for sampling the Brillouin zone. SOC was also considered in this calculation. In order to explore the dynamical stability of the VSSe monolayer, phonon dispersion was obtained by using the finite displacement method²³ as implemented in the Phonopy code.²⁴ Ab initio molecular dynamics (AIMD) simulations with a canonical ensemble were performed to evaluate the thermodynamic stability, and the electric polarizations were computed by using the Berry phase method.²⁵

The particle-swarm search method with three atoms in a hexagonal unit cell predicts the Janus VSSe monolayer to be in the 1T and 2H phases (see Figure 1a,c). Both phases adopt the C_{3v} symmetry, and the lattice constants/total energies are calculated to be 3.25 Å/−19.24 eV and 3.24 Å/−19.31 eV for 1T and 2H VSSe monolayers, respectively. The VSSe monolayer is magnetic, and the 2H phase is more stable than the 1T phase. The magnetic ground state of VSSe monolayer is searched by comparing the energies of various possible magnetic orders, including the nonmagnetic, collinear

ferromagnetic, ferromagnetic, antiferromagnetic, and 120° noncollinear antiferromagnetic orders as illustrated in ref 26 and are shown in Table s1. These were carried out by adopting a $2 \times 2 \times 1$ supercell with four vanadium atoms. It confirmed that the ferromagnetic state has the lowest energy both for the 1T and 2H phases (Table s1). Figure 1b,d presents the calculated phonon spectrum for the 1T and 2H phases, respectively. Clearly, there is no imaginary frequency, indicating high dynamical stability.²⁷ AIMD simulations were further carried out for 10 ps at 300 and 1000 K as shown in Figure s1. These show no evidence of structural destruction, suggesting a robust thermal stability for monolayer VSSe. It is important to note that the Janus MoSSe monolayer has been recently fabricated by using CVD methods.^{6,28} Thus, the experimental realization of a magnetic Janus VSSe monolayer is expected to be straightforward by using a similar approach.

After confirming the stability of magnetic Janus VSSe monolayers at elevated temperatures, we further move to verify if the magnetic field can survive at room temperature. The Curie temperature T_C can be estimated by mapping the DFT total energy over a Heisenberg model and then by using a mean-field approximation (MFA).^{29,30} T_C thus is written as $T_C = \frac{2}{3 * K_B} J$, where J is the exchange coupling parameter (more details can be found in the Supporting Information). Here, we mainly focus on the study of the 2H monolayer VSSe, as it is the most energetically stable. The calculated T_C for 2H VSSe is around 346 K from the PBE method and 1079 K from the HSE method, which is much higher than the transition temperatures observed in exfoliated 2D ferromagnetic CrI_3 .³¹

$$H = \begin{bmatrix} \frac{\Delta}{2} + \epsilon - m_{\text{VB}} & t_{12}(\tau k_x - i k_y) & 0 & 0 \\ t_{12}(\tau k_x + i k_y) & -\frac{\Delta}{2} + \epsilon + \tau\lambda - m_{\text{CB}} & 0 & 0 \\ 0 & 0 & \frac{\Delta}{2} + \epsilon + m_{\text{VB}} & t_{12}(\tau k_x - i k_y) \\ 0 & 0 & t_{12}(\tau k_x + i k_y) & -\frac{\Delta}{2} + \epsilon - \tau\lambda + m_{\text{CB}} \end{bmatrix} \quad (1)$$

where Δ is the bandgap at the valleys (K and K'), ϵ is the correction energy relevant to the Fermi energy, and t_{12} is the effective nearest neighbor hopping parameter. Here, $\mathbf{k} = \mathbf{k} - \mathbf{K}$ is the momentum vector. The SOC-induced spin splitting at CBM (VBM), $2\lambda_{\text{CB}}(2\lambda_{\text{VB}})$, is defined by the energy difference $E_{\text{CB(VB)}}^{\uparrow} - E_{\text{CB(VB)}}^{\downarrow}$ at the K and K' points. In eq 1, $-m_{\text{CB}}(-m_{\text{VB}}) = E_{\text{CB(VB)}}^{\downarrow} - E_{\text{CB(VB)}}^{\uparrow}$ represents the effective exchange splitting at the band edge of the CB (VB). Thus, the exchange interaction, resembling an intrinsic magnetic field, split the spin-up and spin-down states. The four bands (which are contributed by the vanadium orbitals $d_{xy}/d_{x^2-y^2}$ spin-up, d_{z^2} spin-up, $d_{xy}/d_{x^2-y^2}$ spin-down, and d_{z^2} spin-down) are fitted at the K and K' points near the Fermi energy from the DFT results and are given in Figure s4b. By incorporating SOC and the exchange interaction in the $\mathbf{k} \cdot \mathbf{p}$ model, the valley polarization is around 83 meV at the VBM, which is close to the HSE results. Thus, our $\mathbf{k} \cdot \mathbf{p}$ model captures well the valley polarization in the magnetic Janus VSSe monolayer due to the strong coupling between the spin and valley degrees of freedom.

As shown in Figure 1, the length of the V–S chemical bond is shorter than that of the V–Se bond, and the angles between S–V–S and Se–V–Se are different (more details in Figure s5). The anisotropic bonding shears the structure and lowers

and CrGeTe_3 .³² Thus, an intrinsic magnetic field in the 2H VSSe monolayer could persist around room temperature.

Then, the electronic properties of the 2D magnetic Janus VSSe monolayer were accurately calculated by a hybrid functional method.³³ The spin-resolved band structures obtained with and without SOC for the stable 2H and the metastable 1T phases are presented in Figures 2 and s2, respectively. In addition, calculations based on the standard PBE functional (see Figure s3a,b) were also carried out for comparison. Direct spin-up and spin-down gaps are predicted by HSE to be 0.96 and 1.51 eV for the 2H VSSe monolayer (Figure 2a). Interestingly, a valley polarization is present at the K and K' points (see Figure 2b) when SOC is included. The valley energy difference between the K and K' points is around 85 meV at the valence band maximum (VBM), which is mainly contributed by the vanadium $d_{xy}/d_{x^2-y^2}$ orbitals (see red circle in Figure 2c). In contrast, the conduction band minimum (CBM) is dominated by the vanadium d_{z^2} orbital (see green circle in Figure 2c).

The valley polarization can be further understood with a simplified $4 \times 4 \mathbf{k} \cdot \mathbf{p}$ model.³⁴ The basic functions are chosen to be $|\Psi_{\text{CB}}^{\tau}\rangle = |d_{z^2}\rangle$ and $|\Psi_{\text{VB}}^{\tau}\rangle = (|d_{x^2-y^2}\rangle + i\tau|d_{xy}\rangle)/\sqrt{2}$. Carriers in the K and K' valleys in inversion-asymmetric systems belong to opposite Berry curvatures;²⁰ thus, $\tau = \pm 1$ represents the valley index of the K and K' points (more details can be found in the Supporting Information). By incorporating nearest-neighbors hopping parameter t_{12} , the Hamiltonian can be written as

its symmetry, which renders spontaneous deformation in 2H VSSe. In order to explore this effect, we further investigated the magnetic Janus VSSe monolayer in the presence of mechanical strain. Figure 3 presents the change of energy as a function of the uniaxial strain in the 2H magnetic Janus VSSe monolayer.

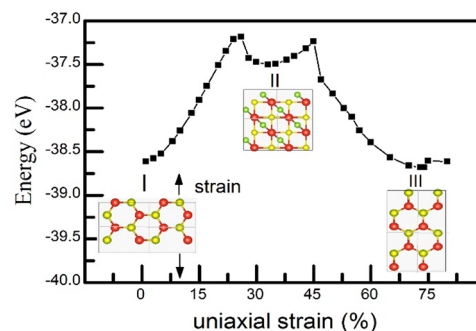


Figure 3. Pathway of ferroelastic switching (I–II–III) for the 2H VSSe monolayer, where yellow, red, and green spheres denote S, V, Se atoms, respectively. The energy profiles of ferroelastic switching as a function of uniaxial strains in the b (zigzag) direction. The inset shows a top view of the $2 \times 2 \times 1$ supercell structure of the VSSe monolayer.

Clearly, there is a spontaneous deformation induced by the external strain, suggesting an intrinsic ferroelasticity. As shown in Figure 3, the initial state I displays parallel armchair and zigzag rows oriented along the a and b directions, respectively. The lattice constant a in the armchair direction is larger than b in the zigzag direction (Table S2). When a uniaxial strain is imposed along the b (zigzag) direction and the lattice parameter along a (armchair) is fully relaxed, the Janus VSSe monolayer transforms into the configuration III, which is energetically more favorable. The lattice parameters (a and b) are exchanged, and the parallel zigzag rows are now along the a direction in state III. This is similar to the initial structure I with a 90° rotation. The configuration labeled as II presents a square lattice, and it represents the intermediate state between I and III, with optimized lattice constants $a' = b' = 4.31 \text{ \AA}$ (Table S2). As a result of the symmetry, the reaction paths from going II to I and from II to III have nearly the same energy barrier. This is calculated to be approximately 0.23 eV/atom in 2H VSSe, a value that is low enough to make room-temperature ferroelasticity possible.¹⁶ Another key indicator on the ferroelastic performance of the compound is the strength of the reversible ferroelastic strain that controls the signal intensity. This is defined as $(\frac{b}{a} - 1) \times 100\%$. The calculated reversible ferroelastic strain for the magnetic Janus 2H VSSe monolayer is as high as 73%. It should be noted that the highest reversible ferroelastic strain up to date among 2D materials is 37.9% in phosphorene.¹⁶ The enormous reversible strain in the magnetic Janus 2H VSSe monolayer indicates an extremely strong switching signal of switching.

Magnetic Janus VSSe possesses crystal asymmetry, suggesting it to be piezoelectric. The piezoelectric effect is the coupling between electrical polarization (P_i) and the strain (ε_{jk}) tensor described by the third-rank tensor $e_{ijk} = \partial P_i / \partial \varepsilon_{jk}$, where i, j , and k correspond to the x, y , and z Cartesian directions.⁸ Symmetry analysis suggests that the magnetic Janus VSSe monolayer (space group C_{3v}) should display two nonvanishing independent piezoelectric coefficients. DFT calculations are then performed with respect to the unstrained state, which is used as a reference. A uniaxial strain is applied along the armchair direction up to 1.5%, and the polarization is estimated by using the Berry phase method²⁵ for both the unstrained and strained cases. Figure 4 presents the in-plane (e_{11}) and out-of plane (e_{13}) piezoelectric coefficients of the monolayer VSSe as a function of the uniaxial strain along the armchair direction. The value of e_{11} for magnetic Janus VSSe is calculated to be $3.303 \times 10^{-10} \text{ C/m}$, which is much larger than that measured for the MoS_2 monolayer, namely, e_{11} of $2.900 \times 10^{-10} \text{ C/m}$.^{8,10} The estimated out-of plane piezoelectric coefficient e_{13} in the 2H VSSe monolayer is about $0.948 \times 10^{-10} \text{ C/m}$, which is relatively weaker than the in-plane coefficient. It should be noted that only rarely 2D materials have out-of plane piezoelectric polarization.

In conclusion, by using first-principles and particle-swarm search approaches complemented with a $k \cdot p$ model, we have predicted a new magnetic Janus monolayer VSSe, which can persist at room temperature. The broken space inversion guarantees inequivalent Berry curvatures. Spin-valley coupling leads to a time-reversal symmetry violation; thus the valley polarization can be realized in VSSe. Moreover, ferroelasticity with a huge reversal strain (73%) also presents in the material. Most interestingly, the coupling between electric polarization and mechanical strain can result in a giant in-plane

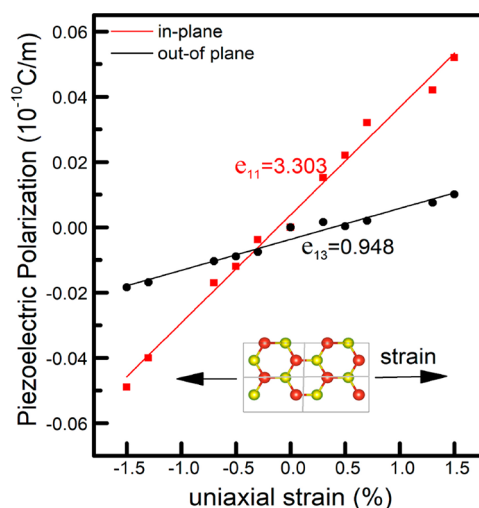


Figure 4. Linear changes in the in-plane and out-of-plane piezoelectric polarizations of the 2H VSSe monolayer under uniaxial strain (armchair) between -1.5 and 1.5% . The slope of the curves gives us the components e_{11} and e_{13} of the stress tensor (unit: 10^{-10} C/m). The inset shows the top view of the $2 \times 2 \times 1$ supercell structure of the 2H VSSe monolayer.

piezoelectric polarization. Our results point to a new interesting magnetic 2D Janus VSSe monolayer that can combine ferromagnetism, ferroelasticity, piezoelectricity, and valley polarization, offering advantages over other 2D materials for potential applications in nanoelectronics, optoelectronics, and valleytronics.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.nanolett.8b05050.

Magnetic ground state and magnetic moment for 1T and 2H VSSe monolayers; method to calculate Curie temperature; MD simulations of 2H and 1T VSSe monolayers under 300 and 1000 K; band structure of 1T and 2H VSSe monolayers; more details on $k \cdot p$ model fitting with DFT results; the optimized lattice constants under strain for the 2H VSSe monolayer; the well-defined polarization along the armchair direction for the VSSe monolayer (PDF)

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Notes

The authors declare no competing financial interest.

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